



Reg.No.:

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JERUSALEM COLLEGE OF ENGINEERING, CHENNAI- 600100
(An Autonomous Institution, Affiliated to Anna University, Chennai)
Academic Year 2025-2026 (Even Semester)

Assessment- I
Fourth Semester

Common to
B.E CSE, B.Tech CS&BS, B.Tech IT, B.Tech AI&DS, B.E CSE (CS), B.E CSE (AIML)

JCS2401 – DATABASE MANAGEMENT SYSTEMS
(Regulation 2023)

Time:08:30a.m.- 10:00a.m.

Date:09-02-2026

Max.Marks:50

PART A (5×2=10marks)

Answer all the Questions

S.No.	Questions	Course Outcome	Bloom's Taxonomy level
1.	Differentiate between Static and Dynamic SQL.	CO1	K1
2.	Define Instance and Schema	CO1	K1
3.	Define Super Key and Primary Key.	CO2	K1
4.	Name and define the two types of participation constraint.	CO2	K1
5.	List ACID properties	CO3	K1

PART B (2×16=32 marks)

6(a)	Explain in detail the basic architecture of a database management system with a neat diagram. (16)	CO1	K2
(OR)			
6(b)	(i) Explain in detail any four Relational Algebra operations with suitable examples.(8) (ii) Explain with proper syntax and examples DML commands(8)	CO1	K2
7(a)	Construct an ER diagram for an University database and complete the ER to Relational Mapping for the same. (16)	CO2	K1
(OR)			
7(b)	Explain each of the following with example. i) Third Normal Form (4) ii) Boyce-Codd Normal Form (4) iii) Fourth Normal Form (4) iv) Fifth Normal Form (4)	CO2	K1

PART C (1×8=8 marks)

8(a)	Consider the following schedules. The actions are listed in the order they are scheduled, and pre-fixed with the transaction name. S1: T1:R(X), T2:R(X), T1:W(Y), T2:W(Y), T1:R(X), T2:R(Y) S2: T3:W(X), T1:R(X), T1:W(Y), T2:R(Z), T2:W(Z), T3:R(Z) For each of the schedules, answer the following questions: (i) What is the precedence graph for the schedule? (ii) Is the schedule conflict_serializable? If so, what are all the conflict equivalent serial schedules? (iii) Is the schedule View_serializable? If so, what are all the view equivalent serial schedules? (16)	CO3	K2, K4
(OR)			
8(b)	Discuss in detail basic Two-phase locking protocol with appropriate example. (16)	CO3	K2